DO NOW — RECALL

# Half adders and full adders

Name: _____ Class: _____ Date: _____

**1. When we group four neighbouring 1s on a Karnaugh map, what happens to the Boolean expression?**

**2. Represent −13 using 8-bit sign and magnitude.**

**3. Insert 40, 20, 60, 10, 30, 50 into an empty binary search tree. Draw the resulting tree.**

**4. Using a 5-bit two’s complement mantissa and 4-bit two’s complement exponent, calculate A − B. Show exponent alignment and mantissa subtraction. A = $0.1100_2 \times 2^3$; B = $0.1000_2 \times 2^2$.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

The Boolean expression becomes simpler. Variables that change within the group can be removed, leaving only variables that stay the same.

### 2. Model answer

Step 1: 13 in binary is 1101. Step 2: pad the magnitude to seven bits: 0001101. Step 3: because the number is negative, put 1 in the sign bit. Answer: 1 | 0001101 = 10001101. This is sign and magnitude , not two’s complement. SIGN · 1 BIT 1 Negative MAGNITUDE · 7 BITS 0001101 13 in binary

### 3. Model answer

Insert the values in the order given . 40 becomes the root. 20 goes left of 40; 60 goes right of 40. 10 goes left of 20; 30 goes right of 20; 50 goes left of 60. Every smaller value goes to the left and every larger value to the right. 40 20 60 10 30 50

### 4. Model answer

Align exponents: B = $0.0100_2 \times 2^3$. Subtract mantissas: $0.1100_2 - 0.0100_2 = 0.1000_2$. Answer: $0.1000_2 \times 2^3 = 4$.

**By the end, you can…**

• derive the half-adder Sum and Carry expressions

• complete half-adder and full-adder truth tables

• explain how full adders connect to add multi-bit binary numbers

**Vocabulary — know these terms**

Recall the term — half adder

Recall the term — full adder

Recall the term — Sum

Recall the term — Carry

Recall the term — carry-in

Recall the term — carry-out

Recall the term — XOR

Recall the term — AND

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**For a half adder, Sum is produced by XOR and Carry is produced by AND.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





NEW LEARNING

# Read, annotate and explain

**Start with binary addition**

One binary column can produce a sum bit and a carry bit.

For 1 + 1, the right-hand bit is Sum = 0 ; the left-hand bit is Carry = 1 .

A half adder adds two single-bit inputs: A and B.

XOR calculates Sum; AND calculates Carry.

Sum = A ⊻ B (XOR) · Carry = A ∧ B

Test every possible pair of inputs.

What happens when a previous column sends a carry into this column?

The half adder only accepts A and B . It has no third input for Carry-in ($C_{in}$) .

A full adder accepts A, B and $C_{in}$.

Add A and B to produce an intermediate Sum $S_1$ and Carry $C_1$.

Feed $S_1$ and Carry-in into a second half adder. Its Sum is the final Sum.

Feed $C_1$ and $C_2$ into an OR gate to obtain Carry-out.

**Start with binary addition**

One binary column can produce a sum bit and a carry bit.

**Meet the half adder**

A half adder adds two single-bit inputs: A and B.

**Discover the two gates**

XOR calculates Sum; AND calculates Carry.

**Build its truth table**

Test every possible pair of inputs.

**Find the limitation**

What happens when a previous column sends a carry into this column?

**First half adder**

Add A and B to produce an intermediate Sum $S_1$ and Carry $C_1$.

**Second half adder**

Feed $S_1$ and Carry-in into a second half adder. Its Sum is the final Sum.

**Combine both carries**

Feed $C_1$ and $C_2$ into an OR gate to obtain Carry-out.

**Completed full adder**

Trace both routes: Sum through the second half adder; Carry-out through OR.

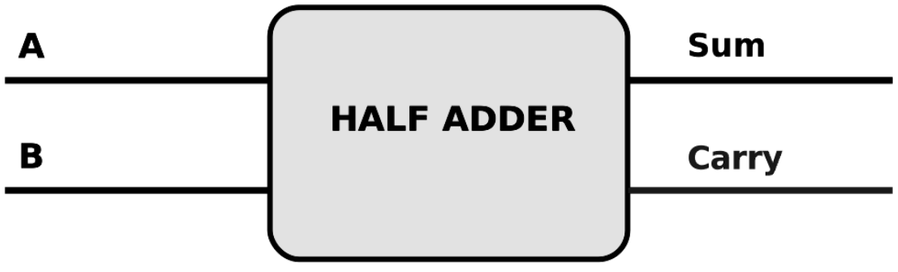
**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Meet the half adder

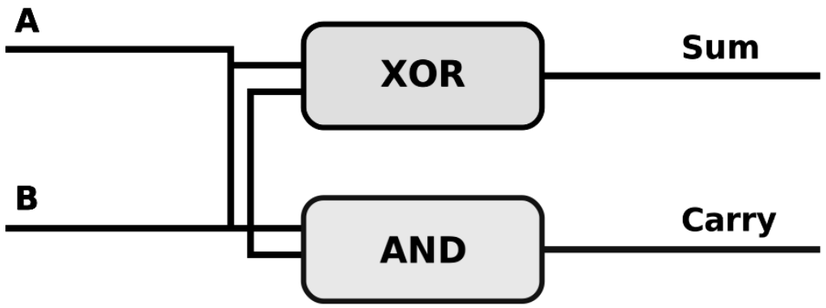
A half adder adds two single-bit inputs: A and B.




Half adder has inputs A and B and outputs Sum and Carry

### Discover the two gates

XOR calculates Sum; AND calculates Carry.




Half adder: A and B enter both XOR and AND gates





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain four-bit addition
Explain how four full adders can add two 4-bit binary numbers.

### Question 4

Add 0111 + 0001
Use bit-by-bit addition and identify where the carry travels.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Half-adder question
State the Sum/Carry expressions, complete the truth table, then explain how half adders contribute to a full adder.

### Question 2

Complete eight cases
Without looking at the diagram, sketch the five gates and their connections from memory. Then create the full-adder truth table for A, B and CarryIn.



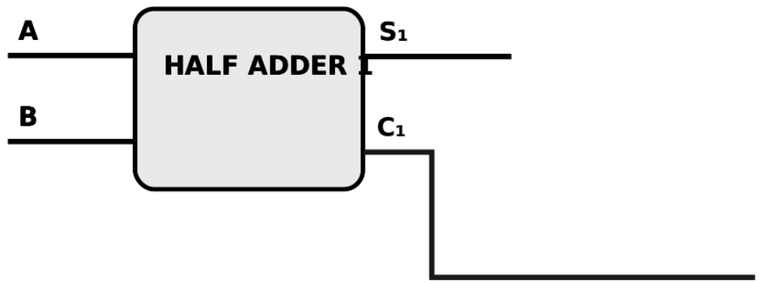


# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### First half adder · 1 of 3

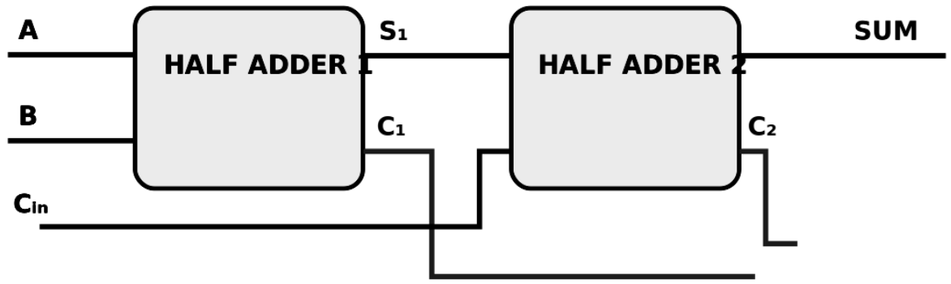
Add A and B to produce an intermediate Sum $S_1$ and Carry $C_1$.




Full adder built from two half adders and an OR gate

### Second half adder · 2 of 3

Feed $S_1$ and Carry-in into a second half adder. Its Sum is the final Sum.



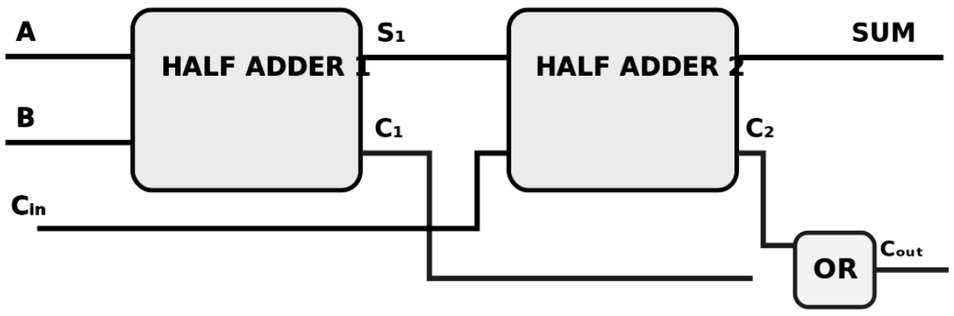

Full adder built from two half adders and an OR gate

## New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Connect both carries · 3 of 3

Feed $C_1$ and $C_2$ into an OR gate to obtain Carry-out.




Full adder built from two half adders and an OR gate



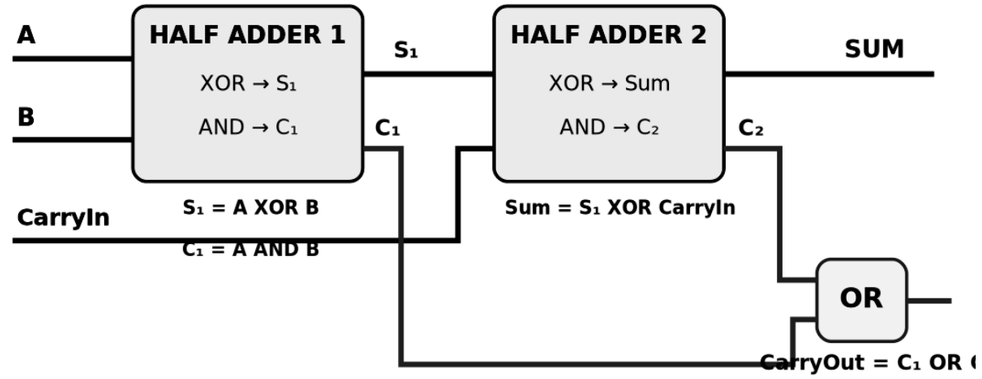


## We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### From half to full adder

Trace the two XOR gates, then the two AND gates feeding OR.




Full adder built from two half adders: A and B enter the first half adder; its sum S1 and CarryIn enter the second half adder; carry C1 and C2 are combined using an OR gate to produce CarryOut

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

### 1. Half-adder expressions

OCR asks for S and C.

```
S = A ? B
C = A ? B
```

Method / working:

### 2. Half-adder truth table

Complete the four outputs.

```
A B | S C
0 0 | ? ?
0 1 | ? ?
1 0 | ? ?
1 1 | ? ?
```

Method / working:

### 3. From half to full adder

Trace the two XOR gates, then the two AND gates feeding OR.

```
Two half adders plus an ? gate can form a full adder.
```

Method / working:

### 4. Full-adder case

A=1, B=1, CarryIn=1.

```
1 + 1 + 1 = binary ?
Sum = ?
CarryOut = ?
```

Method / working:

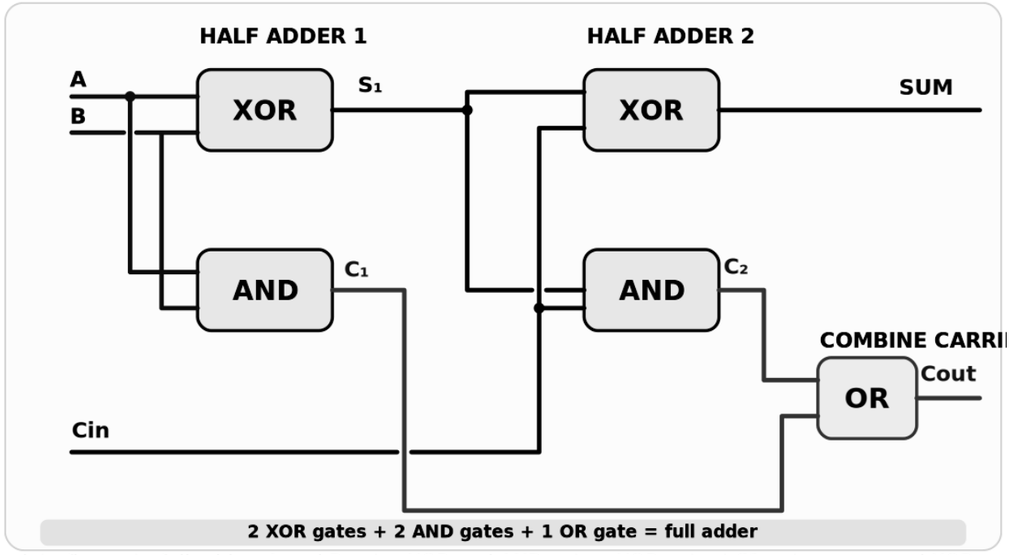




## New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Full Adder — All Five Gates




Complete five-gate full adder. A and B enter XOR and AND gates. XOR output S1 enters a second XOR and AND with Carry-in from the left. The second XOR outputs Sum; both AND carry outputs C1 and C2 enter OR and produce Carry-out. Crossed wires without junction dots are not connected.

I DO — WORKED EXAMPLE

# Follow the worked method

### Model 1 + 1 with a half adder

### Rightmost result bit

```
Sum = 0
```

### Next column

```
Carry = 1
```

### Trace XOR and AND

```
1 XOR 1 = 0 → Sum
1 AND 1 = 1 → Carry
```

### Connect both half adders

```
S₁ = A XOR B
Sum = S₁ XOR Cin
Cout = (A AND B) OR (S₁ AND Cin)
```



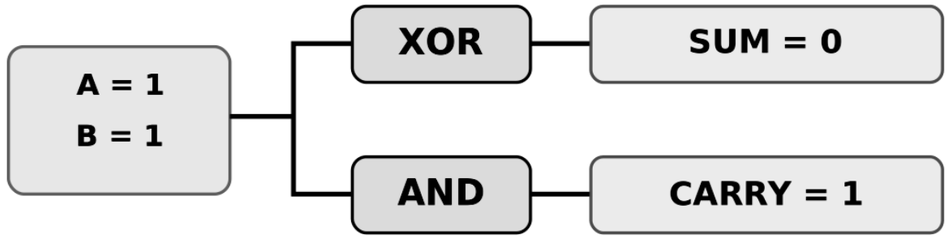


# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Model 1 + 1 with a half adder

We have just learned XOR and AND. Now model the output when A = 1 and B = 1.




**1 + 1 = 10₂ → sum bit 0, carry bit 1**

Half adder with inputs A and B both one; XOR produces sum zero and AND produces carry one